KE YANG

Department of Computer Science
The University of Texas at San Antonio
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CURRENT AFFILIATIONS

Assistant Professor
Department of Computer Science,
College of AI, Cyber, and Computing,
The University of Texas at San Antonio — Aug, 2023 - current

Founding Faculty
Cohort for AI REsponsibility (CAREAI),
The University of Texas at San Antonio — Sep, 2023 - current

Core Faculty
Center for Data Science,
The University of Texas at San Antonio — Sep, 2023 - current

Core Faculty
AI Consortium for Human Well-being (MATRIX),
The University of Texas at San Antonio — Oct, 2023 - current

PAST ACADEMIC APPOINTMENTS

Postdoctoral Research Associate
Manning College of Information & Computer Sciences,
University of Massachusetts Amherst — Sep, 2021 - Aug, 2023

Postdoctoral Fellow
Center for Data Science,
University of Massachusetts Amherst — Sep, 2021 - Aug, 2023

Visiting Scholar
Tandon School of Engineering,
New York University — 2021 - 2022

RESEARCH INTEREST

My work is broadly in trustworthy AI and ML, data management, and human-centered data science, aiming to develop data-centric solutions in knowledge-intensive fields, including genomics, medicine, healthcare, finance, and legal, etc. My research lies at the intersection of data and predictive models (e.g., using Artificial Intelligence (AI) and Machine Learning (ML) algorithms), with a focus on *hallucinations*, *explainability*, *algorithmic fairness*, and *the mitigation of unreliable solutions*. I also work in enhancing *AI and ML education* and public engagement.

EDUCATION

Ph.D. in Computer Science, New York University 2019 - 2021
- Advisor: Julia Stoyanovich, GPA 3.9/4.0
- Thesis: Fairness, diversity, and interpretability in ranking
- Committee: Julia Stoyanovich (chair), Daniel Neill, Oded Nov, and Sebastian Schelter

Ph.D. candidate in Computer Science, Drexel University (transferred) 2015 - 2018
- Advisor: Julia Stoyanovich, GPA 3.98/4.0

M.S. in Computer Science, Beijing Technology and Business University, 2012 - 2015
- Advisors: Qian Mo and Zhongming Han, GPA 3.51/4.0
- Thesis: Detecting web spammers in online shopping websites
- Graduated *summa cum laude*

B.Eng. in Software Engineering, Beijing Technology and Business University 2008 - 2012
- Advisor: Qian Mo, GPA 3.45/4.0
- Graduated *summa cum laude*

GRANTS, FELLOWSHIPS, AWARDS, & HONORS

- Connect & Thrive Mini-Grant ($1,000) for Spring 2026, Student Success Program, UT San Antonio, December 2025
- Reducing Hallucination of LLMs for Causality-integrated Personal Medical History Question Answering, Collaborative Seed Funding Grant ($35,000), School of Data Science (SDS) and the Open Cloud Institute (OCI), UTSA, 2024
- Faculty Travel Support Program, Internal funding, Office of Research, UTSA, April 2024
- Sponsored participant for NSF Secure and Trustworthy Cyberspace program (SaTC) Aspiring PI Workshop, Chicago, May 2024
- Sponsored faculty for NCFDD Faculty Success Program summer 2024, Office of Research, UTSA
- UTSA funding for computing equipment ($250,000), STARs Program, the University of Texas System, 2023
- Postdoctoral Fellowship, *Center for Data Science*, University of Massachusetts Amherst, 2021

– Pearl Brownstein Doctoral Research Award, *Tandon School of Engineering*, New York University, 2021

PUBLICATIONS

Peer-reviewed Conference Publications

1. Asif Ahmed, Sakhawat Hossain Saimon, Jianhua Ruan, and **Ke Yang**. 2026. NiWo: An Augmentation Framework to Enhance ML Performance and Interpretability for Tabular Data with Class Imbalance. In Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26), August 09–13, 2026, Jeju Island, Republic of Korea. ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3770855.3817894.
2. Sinmi Akindele, Youmi Suk, **Ke Yang**. An Empirical Study of Synthetic Data Generators for Sequences. In Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26), August 09–13, 2026, Jeju Island, Republic of Korea. ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/3770855.3817521.
3. Erfan Nourbakhsh, Rocky Slavin, **Ke Yang**, Anthony Rios. When Retrieval Doesn't Help: A Large-Scale Study of Biomedical RAG. Biomedical Natural Language Processing Workshop (BioNLP) at ACL 2026, San Diego, CA, USA.
4. Md Sohanur Rahman, Anthony Rios, and **Ke Yang**. Understanding Robustness of RAG Paradigm in Medical QA. AIM Health R&D Summit 2026. Present as a Rapid Fire Talk. San Antonio, USA.
5. Issac Alvarez, Victoria Johnson, and **Ke Yang**. Developing Interpretable Vision Question-Answering LLMs for Radiology Images. AIM Health R&D Summit 2026. Present as a poster. San Antonio, USA.
6. Naveen John and **Ke Yang**. Evaluating Bias in LLMs' Responses to Patient-oriented Tasks. AIM Health R&D Summit 2026. Present as a poster. San Antonio, USA.
7. Kevin Mathew Maran, Kevin Hirschi, and **Ke Yang**. Evaluating Automatic Speech Recognition Models for Bias from Speaker Characteristics, Context, and Linguistic Dimensions. AIM Health R&D Summit 2026. Present as a poster. San Antonio, USA.
8. Victoria Johnson, Issac Alvarez, and **Ke Yang**. Assessing the capabilities of Vision Question-Answering LLMs in Understanding Radiology Images. AIM Health R&D Summit 2026. Present as a poster. San Antonio, USA.
9. Youmi Suk, Chenguang Pan, and **Ke Yang**. Using Generative AI for Sequential Data Generation in Monte Carlo Simulation Studies. Lecture-style presentation at the National Council on Measurement in Education (NCME 2026) Annual Meeting, Los Angeles, California, USA.
10. Đorde Klisura, Astrid R. Bernaga Torres, Anna Karen Gárate-Escamilla, Rajesh Roshan Biswal, **Ke Yang**, Hilal Pataci, and Anthony Rios. A Multi-Agent Framework for Mitigating Dialect Biases in Privacy Policy Question-Answering Systems. ACL main conference 2025, Vienna, Austria, arXiv preprint
11. Md Sohanur Rahman, Yuexia Zhang, Anthony Rios, and **Ke Yang**. Extracting Causal Relational Rules for Medical Question-Answering Tasks using Large Language Models. Workshop on Generative AI for Smart and Connected Health: Innovations, Challenges, and Applications (GenAI4SCH),

co-located with IEEE/ACM conference on Connected Health: Applications, Systems and Engineering Technologies **(CHASE 2025)**, New York, USA, regular short paper.

12. Hilal Pataci, **Ke Yang**, and Anthony Rios. Empowering Privacy Policy Understanding and Accessibility in Smart Home Devices. **INFORMS 2024**, presented as a poster.
13. Youmi Suk and **Ke Yang**. Using Conditional Tabular Generative Adversarial Networks for process data generation in Monte Carlo simulation studies. Paper presented at the International Meeting of Psychometric Society (**IMPS 2024**), Prague, Czech Republic.
14. **Ke Yang** and Alexandra Meliou. Non-Invasive Fairness in Learning through the Lens of Data Drift. IEEE International Conference on Data Engineering **(ICDE 2024)**, Utrecht, Netherlands.
15. **Ke Yang**. DEEILS: Data Ethics Embedded Interactive Learning System for Computer Science Students. The Technical Symposium on Computer Science Education **(SIGCSE TS 2024)**, Portland, Oregon, United States, presenting as a lighting talk for junior faculty.
16. **Ke Yang**, Joshua R. Loftus, and Julia Stoyanovich. Causal Intersectionality and Fair Ranking. Symposium on Foundations of Responsible Computing, (**FORC 2021**), Virtual Conference.
17. Meike Zehlike, **Ke Yang**, and Julia Stoyanovich. Mapping Normative Frameworks to Fair Ranking. ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (**EAAMO 2021**), presented as a poster.
18. **Ke Yang**, Vasilis Gkatzelis, and Julia Stoyanovich. Balanced Ranking with Diversity Constraints. International Joint Conference on Artificial Intelligence (**IJCAI 2019**), Macao, China.
19. **Ke Yang**, Julia Stoyanovich, Abolfazl Asudeh, Bill Howe, H.V. Jagadish, and Gerome Miklau. A Nutritional Label for Rankings. ACM International Conference on Management of Data (**SIGMOD 2018 Demonstration**), Houston, USA.
20. Julia Stoyanovich, **Ke Yang**, H.V. Jagadish. Online Set Selection with Fairness and Diversity Constraints. International Conference on Extending Database Technology (**EDBT 2018**), Vienna, Austria.
21. **Ke Yang** and Julia Stoyanovich. Measuring Fairness in Ranked Outputs. International Conference on Scientific and Statistical Database Management (**SSDBM 2017**), Chicago, USA.

Peer-reviewed Journal Publications

19. Youmi Suk, Chenguang Pan, and **Ke Yang**. Using Generative AI for Sequential Data Generation in Monte Carlo Simulation Studies, Journal of Educational and Behavioral Statistics, DOI: 10.3102/10769986251397559, 2025.
20. Zehlike Meike, **Ke Yang**, and Julia Stoyanovich. Fairness in Ranking, Part I: Score-based Ranking. ACM Computing Surveys (**CSUR 2022**).
21. Zehlike Meike, **Ke Yang**, and Julia Stoyanovich. Fairness in Ranking, Part II: Learning-to-Rank and Recommender Systems. ACM Computing Surveys (**CSUR 2022**).
22. Zhongming Han, **Ke Yang**, and Xusheng Tan. Analyzing Spectrum Features of Weight User Relation Graph to Identify Large Spammer Groups in Online Shopping Websites. **Chinese Journal of Computers**, 40(4): 939-954 (2017) (in Chinese).
23. Zhongming Han, **Ke Yang**, Fengmin Xu, and Dagao Duan. Probabilistic Graphical Model for Detecting Spammers in Microblog Websites. **Journal of Embedded Systems**, 8(1): 12-23 (2016).

24. Qian Mo and **Ke Yang**. Overview of Web Spammer Detection. **Chinese Journal of Software**, 25(7): 1505-1526 (2014) (in Chinese).

Peer-reviewed Workshops

25. Md Sohanur Rahmand, Anudeep Bonagiri, Shreyansh Rath, Anthony Rios, and **Ke Yang**. A Multi-Agent Framework for Faithful Evaluation of Medical Summaries. **Texas NLP Symposium 2026**, College Station, TX, USA.
26. Veronica Rammouz, Aaron O. Gonzalez, Carlos Cruzportillo, Adrian Tan, Nicole Lang Beebe, **Ke Yang**, and Anthony Rios. Absolute Error Is Not Enough: Ranking Model Performance Without Labels. **Texas NLP Symposium 2026**, College Station, TX, USA.
27. Xingmeng Zhao, **Ke Yang**, and Anthony Rios. Comparing Clinical and General LLMs on Knowledge Boundaries and Robustness. Mechanistic Interpretability Workshop at NeurIPS, **(Mech Interp Workshop@NeurIPS 2025)**, San Diego, CA, USA.
28. **Ke Yang**, Biao Huang, Julia Stoyanovich, and Sebastian Schelter. Fairness-Aware Instrumentation of Preprocessing Pipelines for Machine Learning. Workshop on Human-In-the-Loop Data Analytics in conjunction with the 2020 ACM SIGMOD/PODS Conference, (**HILDA@SIGMOD 2020**), Portland, OR, USA.

INVITED TALKS

- Utilizing LLMs to Facilitate Medical and Healthcare Applications, Mays Cancer Center / UT Health San Antonio Survivorship Retreat, March 2026
- Data-driven Solutions for Trustworthy AI, *Keynote Speaker*, Alamo Regional Data Alliance (ARDA) 2025 Spark Conference: The Power of Data Ethics
- AI Agents for Online Information Summarization and Explanation, AI Workshops Month, ACM San Antonio, April 2025
- Reducing Hallucinations of LLMs for Medical Question Answering, the 2nd Annual Los Datos Conference @ UTSA, October 2024
- Data-driven Solutions for AI Responsibility, Virtual Seminar, DATA Lab @ Northeastern University, 2024 Spring
- AI and the Changing Nature of Work, *Panel Speaker*, South Texas Human Resources Symposium (STXHRS) 2024
- Causal Intersectionality and Fair Ranking, Integrated Symposium on Algorithmic and Testing Fairness in Education: Methods and Applications, SREE 2023 Conference Embedding Equity for Better Methods, Evidence, and Practice, September 2023
- Mitigating Bias in Data-driven Tasks, UTSA MATRIX, Fall Seminar Series, September 2023
- Responsible Data Science, Brown University, virtual, March 2023
- Reducing the Impact of Bias on Data-driven Tasks, Ohio State University, February 2023
- Data Science for Social Good, *Panel Speaker*, Central Massachusetts Women in Data Science (WiDS), Worcester Polytechnic Institute (WPI), March 2022

- Data Bias and their Impact on Machine Learning, MIT Horizon event, November 2022
- Fairness and Diversity in Rankings, AT&T Lab Research Weekly Symposium, June 2019
- Fairness In Ranking Systems, AT&T Lab Graduate Student Symposium, November 2018

TEACHING

Instructor at the University of Texas at San Antonio

- CS 6283 *Deep Learning*, Spring 2026
- CS 5823 *Trust, Confidence, and Explainability in AI*, Spring 2025, 2026
- CS 7123 *Research Methods*, Spring 2024
- CS 5443 *Database Management Systems*, Fall 2023, 2024, 2025

Assisting Instructor at University of Massachusetts Amherst

- CMPSCI 345 *Practice and Applications of Data Management*, Fall 2022

Teaching Assistant at Drexel University

- CS 500 *Fundamentals of Databases*, Winter 2017
- CS 461 *Database Systems*, Spring 2017

STUDENTS' MENTORING

Current

- Sinmi Akindele, Ph.D. student, University of Texas at San Antonio, Computer Science, joined in Fall 2023
- Sohanur Rahman, Ph.D. student, University of Texas at San Antonio, Computer Science, joined in Fall 2023
- Asif Ahmed, Ph.D. student, University of Texas at San Antonio, Computer Science, joined in fall 2023
- Kevin Mathew Maran, Master's student, University of Texas at San Antonio, Computer Science, joined in fall 2024
- Naveen John, Master's student, University of Texas at San Antonio, Computer Science, joined in spring 2025
- Issac Alvarez, Master's student, University of Texas at San Antonio, MSAI, joined in fall 2025
- Abhishek Patil, Master's student, University of Texas at San Antonio, Computer Science, joined in fall 2025

- Julian Beltz, Master's student, University of Texas at San Antonio, Computer Science, joined in fall 2025
- Jonathan Gladwin Prabakar, undergraduate student, University of Texas at San Antonio, Computer Science, joined in fall 2025
- Anudeep Bonagiri, undergraduate student, University of Texas at San Antonio, Computer Science, joined in spring 2025
- Victoria Johnson, undergraduate student, University of Texas at San Antonio, Computer Science, joined in fall 2025
- Anh Doan, undergraduate student, University of Texas at San Antonio, Computer Science, joined in spring 2026
- Bhargav Ratala, undergraduate student, University of Texas at San Antonio, Computer Science, joined in fall 2025
- Muhammad Baibars Zain Ul Abideen, undergraduate student, University of Texas at San Antonio, Computer Science, joined in fall 2025
- Phu Pham, undergraduate student, University of Texas at San Antonio, Computer Science, joined in spring 2026
- Shreyansh Rath, high school student, Heritage High School, joined in spring 2025
- Steawin Igwin Fernando, high school student, BASIS San Antonio, Shavano Campus, joined in spring 2026

Past

- Michael De La Rosa, Master's student, University of Texas at San Antonio, MSAI, participated in spring 2025
- Gairo Yostin Peralta Álvarez, Master's student, Instituto Tecnológico de Estudios Superiores de Monterrey (TEC), participated in summer 2025
- Colette Atupulazi, Master's student, University of Texas at San Antonio, Computer Science, participated in fall 2025 and spring 2026
- Sai Siri Chittineni, undergraduate student, University of Texas at San Antonio, Computer Science, participated from fall 2023 through fall 2025, supported by SDS Undergraduate Research Fellowship at UT San Antonio
- Damien Beltran, undergraduate student, University of Texas at San Antonio, Computer Science, participated in fall 2025
- Iván Alejandro Pedraza Macias, Master's student, Instituto Tecnológico de Estudios Superiores de Monterrey (TEC), participated in summer 2025
- Jason Johnson, Master's student, University of Texas at San Antonio, Computer Science, participated in spring and fall 2024
- Samyuktha Arcot Megha Raj, Master's student, University of Texas at San Antonio, Computer Science, participated in fall 2024
- Kosha Antala, undergraduate student, University of Texas at San Antonio, Computer Science, participated in fall 2025

- Ashmit Sethi, high school student, Dublin High School, participated in fall 2024
- Rajarshi Ghosh, high school student, Oak High School and Lone Star College, participated during spring 2024
- Zafeiria Moumoulidou, Ph.D. student, University of Massachusetts Amherst, Manning College of Information & Computer Sciences, co-advised with Prof. Alexandra Meliou in 2022 and 2023
- Alexandra Burushkina, Master's student, University of Massachusetts Amherst, Center for Data Science, Data Science for Social Good Summer Program, graduated June 2023, current position: Ph.D. student in Computer Science at University of Massachusetts Amherst
- Peter Tran, Master's student, University of Massachusetts Amherst, Center for Data Science, Data Science for Social Good Summer Program, graduated June 2023, current position: Ph.D. student in Psychometrics at University of Massachusetts Amherst
- Pragath Siva, high school student, Sharon High School, graduated June 2023, current position: freshman in Computer Science and Economics at the University of Illinois Urbana-Champaign
- Samasth Norway Ananda, Master's student, New York University, graduated June 2021, current position: software developer at Oracle
- Biao Huang, Master's student, New York University, graduated June 2020, current position: software developer in China lead tech company
- Jensine Raihan, undergraduate student, New York University, 2020

ACADEMIC SERVICE

Conference Organization

- Session co-chair: Responsibility and Privacy at Extending Database Technology (EDBT) 2022
- Technical chair: 4th Workshop on Data Management for End-to-End Machine Learning (DEEM) at SIGMOD 2020

Program Committee Member

- EDBT 2027
- AAAI Social Impact Track, 2026
- Workshop on Linguistic Analysis for Health (HeaLing'26), Co-located with EACL, 2026
- Guided AI Workshop @ ACM Conference on Management of Data (SIGMOD) 2025, 2024
- International Conference on Extending Database Technology (EDBT) Industry Program 2023, 2022
- ACM Web Conference (a.k.a. International World Wide Web Conference, abbreviated as WWW) 2026 Industry, 2024 Industry, 2023 Industry, Workshops, Tutorials, Posters, and Demos, 2022 Industry
- IEEE Conference on Data Engineering (ICDE) Demo Track, 2023, 2022
- ACM Conference on Fairness, Accountability, and Transparency (ACM FAccT) 2021

- ACM Workshop on Responsible Recommendation (FAccTRec) at RecSys 2025, 2024, 2023, 2022, 2021
- ACM Workshop on Data Management for End-to-End Machine Learning (DEEM) at SIGMOD 2021, 2020

Reviewer

- KDD 2026
- AAAI 2026
- ACM FAccT 2026
- Journal of Data Mining and Knowledge Discovery, 2026, 2025
- Ethics reviewer, ICML 2026, 2025
- Ethics reviewer, NeurIPS 2025, 2024, 2023, 2022, 2021
- ACM Conference on Management of Data (SIGMOD) 2023
- ACM Transactions on Social Computing 2023
- ACM Conference on Human Factors in Computing Systems (CHI) 2021

Other Activities

- Social Science Panelist for American Association of University Women (AAUW), International Fellowships, 2026, 2025, 2024
- Student volunteer: ACM Conference on Fairness, Accountability, and Transparency (ACM FAT*) 2019

PAST PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate
Manning College of Information & Computer Sciences,
University of Massachusetts Amherst, MA, U.S.A. 2021 - 2023
- Supervisor and co-PI: Alexandra Meliou
- Fully funded by CDS Postdoctoral Fellowship
- Selected publication: Non-Invasive Fairness in Learning through the Lens of Data Drift, 2023

Graduate Research Assistant
VIDA Research Center, New York University, NY, U.S.A. 2019 - 2021
- Supervisor: Julia Stoyanovich
- Fully funded by a graduate research assistantship
- Selected publication: Fairness in Ranking: A Survey, 2021 (received 145 citations until Aug, 2023)
- More of my past projects on dataresponsibly.github.io I built for the lab during graduate school.

Research Intern
AT&T Labs, New York City, NY, U.S.A. 2019

- Supervisors: Emily Dodwell, Ritwik Mitra, and Balachander Krishnamurthy
- Project: Fairness and transparency in machine learning

Graduate Research Assistant
College of Computing & Informatics, Drexel University, PA, U.S.A. 2015 - 2018
- Supervisor: Julia Stoyanovich
- Fully funded by a graduate research assistantship
- Selected publication:
> Measuring Fairness in Ranked Outputs, 2017 (received 341 citations until Aug, 2023)
> A Nutritional Label for Rankings, 2018 (received 113 citations until Aug, 2023)

Research engineer
Elite & Resource (start-up company), Beijing, China 2014 - 2015
- Supervisor: Peng Sun
- Project: Preventing flood disasters using machine learning
- Results have been integrated as a core component of a national data management system for floor disasters

Graduate Research Assistant
Beijing Technology and Business University, Beijing, China 2012 - 2015
- Supervisors: Zhongming Han and Qian Mo
- Fully funded by a graduate research assistantship
- Selected publication:
> Overview of Web Spammer Detection, 2013 (published in a top-tier computer science journal in Chinese)
> Analyzing Spectrum Features of Weight User Relation Graph to Identify Large Spammer Groups in Online Shopping Websites, 2015 (published in a top-tier computer science journal in Chinese)